

Diagnostics

One screen per input or output. Focused content is large. Every detail page names the GPIO and Board A terminal and shows the live level. Supporting values stay small. Header prefix is DIAG ·.

Draft. Pixel layout: UI/diagnostics.html. Terminal numbers match Board A.

Jog and the obstruction test own the motor. They carry an amber banner while live. Hall and rocker cannot preempt them. HOLD cancels: PWM off, current decay, K1 unchanged. Jog is a timed EC11 (or web keypad) nudge. PRESS while idle edits duty (5–25%) and step (200–1000 ms). It does not need the shaft encoder, markers, or current sensor. Leftover queue time expires on its own, so a closed tab cannot leave the motor latched.

1. Menu

Twelve items. Eight fit. TURN windows the list. Pips flag anything already unhealthy. PRESS opens. HOLD exits to Main. Items 9–12 are off this shot. Each has its own figure below.

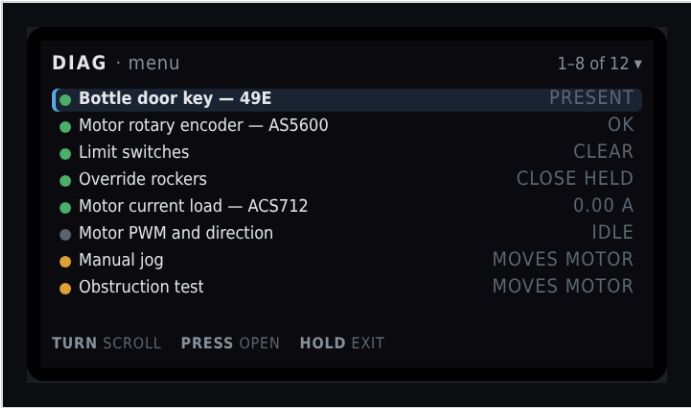


Figure 1. DIAG · menu. First window, bottle key selected.

On screen	Meaning
DIAG · menu	Header left.
1-8 of 12 ▾	Header right. This window is the first eight rows. The triangle means more items below.
Green pip + highlight + Bottle door key — 49E + PRESENT	Selected. Hall sensor. Value is live cooked state.
Green pip + Motor rotary encoder — AS5600 + OK	Shaft magnet detected, I ² C good.
Green pip + Limit switches + CLEAR	Neither marker is made.
Green pip + Override rockers + CLOSE HELD	Close throw is made right now. Centre does nothing. PRESS on that page pulses a throw through the same path as the hardware.
Green pip + Motor current load — ACS712 + 0.00 A	Live amps at rest.
Grey pip + Motor PWM and direction + IDLE	Duty zero. Grey pip is idle, not a fault.
Amber pip + Manual jog + MOVES MOTOR	Amber because opening this page can command the motor. Position is not required.
Amber pip + Obstruction test + MOVES MOTOR	Same warning. Plastic-bottle test. Does not write the learned table.
TURN scroll · PRESS open · HOLD exit	Window the list, enter the highlighted item, or return to Main.

TURN past item 8 to reach:

#	Menu item	Value column	Opens
9	Obstruction table	graphs	Close-direction graph first. TURN swaps to open.
10	EC11 knob and button	live detents	Quadrature, detent scale, display timeout remaining.
11	Network and OTA	STA or AP	Address, mDNS, antenna note.
12	Stored calibration	armed / stale	Which steps are valid and why a stale one failed.

Every detail page keeps a pin strip: GPIO, terminal, function, live level or raw value. HOLD back returns to this menu. Bottle key, limit markers, and override rockers can inject a cooked signal from PRESS. The pin strip stays the real GPIO.

2. Bottle door key — 49E

Present and absent are two calibrated levels with separate enter and exit thresholds. Polarity is stated in words because it is configurable. PRESS can pulse the cooked state without moving the bottle.

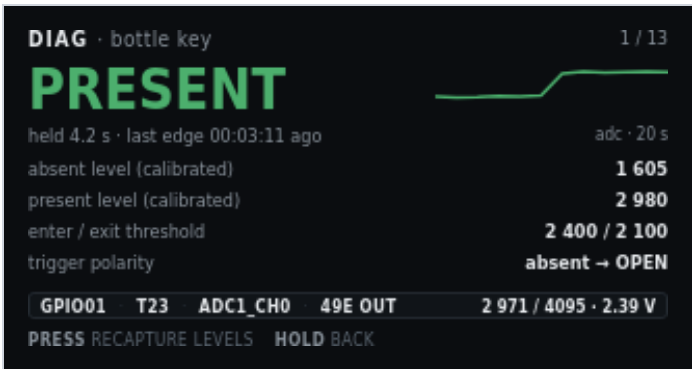


Figure 2. DIAG · bottle key. Present, last edge 3 min 11 s ago.

On screen	Meaning
DIAG · bottle key	Header left.
1 / 13	Header right. First of thirteen detail screens. The menu is not numbered here.
PRESENT (green)	Hero. Cooked state after hysteresis. Amber plus SIM when PRESS has forced the state.
pulse · adc 2971 · last edge ...	Selected action (pulse or recapture), live ADC, and age of the last cooked edge. Sparkline is the real ADC, newest at the right.
Sparkline + adc · 8 s	Raw ADC, newest at the right (dot). About 8 seconds once the strip is full. The trace scrolls left. The step up is the bottle going in.
absent level (calibrated) · 1 605	Stage 1c absent capture.
present level (calibrated) · 2 980	Stage 1c present capture.
enter / exit threshold · 2 400 / 2 100	Hysteresis pair derived from those levels.
trigger polarity · absent → OPEN	Automatic Main command when the key leaves the recess.
Pin strip: GPIO01 · T23 · ADC1_CH0 · 49E OUT	MCU pin, Board A terminal, ADC channel, sensor output name. Always the real pin, even during a pulse.
2 971 / 4095 · 2.39 V	Live raw and volts at the pin.
TURN action	Pulse the key, or recapture levels. Pulse is the default. The highlight stops at each end. Extra clicks do not wrap.
PRESS pulse	Toggle cooked present/absent. Fires the same open/close edge as the hall sensor. Does not fake the ADC.
PRESS recapture	After TURN: capture absent, then PRESS again with the bottle in. Same two-step as Calibration 1c.
HOLD back	Return to the diagnostic menu. Clears a simulated key state.

3. Motor rotary encoder — AS5600

The only source of door position. It also carries the jitter band and the drift from the last marker re-sync.



Figure 3. DIAG · shaft encoder. 62% of span, magnet OK.

On screen	Meaning
DIAG · shaft encoder	Header left.
2 / 13	Header right.
12 690 cnt	Hero. Accumulated counts from the open origin.
62% of span · 3.10 rev from origin	Share of measured span, and shaft revolutions since origin.
Angle dial	Single-turn AS5600 angle inside the current revolution. Not the multi-turn door position.
148.3°	That single-turn angle in degrees.
1.7 rev/s CCW	Live shaft speed and direction.
raw angle · magnet · 1 687 · OK	12-bit raw angle (0–4095) and magnet quality.
AGC · status · 128 · detected	Automatic gain and magnet-detected flag from the chip.
jitter band at rest · ± 3 cnt	Stage 1b capture. Growing band is an early warning for a weak magnet.
drift last re-sync (open / close) · -12 / +7 cnt	Correction applied the last time each marker was seen. Growing drift means lost turns or a slipping magnet.
Pin strip: GPIO43/44 · T10/T9 · I²C 0x36 · 12-bit	SCL/SDA, terminals, bus address, resolution.
4096 cnt/rev · 100 kHz	Counts per mechanical revolution and I²C clock.
PRESS zero drift log	Clear the displayed per-marker drift totals. Does not change origin.
HOLD back	Return to the menu.

4. Limit switches

HIGH is the marker state. A cut wire is indistinguishable from a made switch, so the screen says so. Both markers active is a fault, shown here rather than as a silent refusal to start. PRESS can force a marker without blocking the optical path.

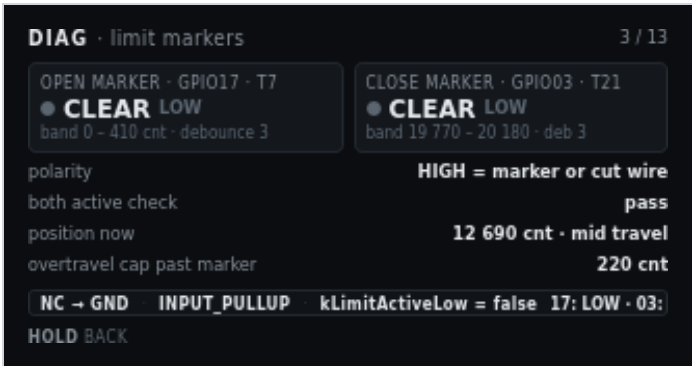


Figure 4. DIAG · limit markers. Both clear, mid travel.

On screen	Meaning
DIAG · limit markers	Header left.
3 / 13	Header right.
Open marker · GPIO17 · T7	Left tile title. Pin and Board A terminal.
Grey pip + CLEAR LOW	Open switch not made. Electrical level is LOW.
band 0 - 410 cnt · debounce 3	Count window where an open-marker actuation is accepted, and debounce samples.
Close marker · GPIO03 · T21	Right tile title.
Grey pip + CLEAR LOW	Close switch not made.
band 19 770 - 20 180 · deb 3	Close-marker acceptance window and debounce.
polarity · HIGH = marker or cut wire	NC to GND with pull-up. HIGH means the loop is open: at the end, or the wire is cut.
both active check · pass	Both markers made at once is a fault. Pass means that is not happening.
position now · 12 690 cnt · mid travel	Live counts and a cooked region name.
overtravel cap past marker · 220 cnt	Hard cap after a marker. A seek that passes this is a fault.
Pin strip: NC → GND · INPUT_PULLUP · kLimitActiveLow = false	Wiring and the firmware polarity constant.
17: LOW · 03: LOW	Live GPIO17 and GPIO3 levels. Stays the real pin during a force.
TURN select · PRESS force	Highlight open or close. The highlight stops at each end. Extra clicks do not wrap. PRESS latches that marker as if the switch were made. Amber pip and SIM on the tile. PRESS again clears that force. A forced destination marker will stop a move already in progress.
HOLD back	Return to the menu. Clears a forced marker.

5. Override rockers

Two paralleled stations act as one logical switch. Each throw is shown separately. A stuck throw is invisible if you only display the combined result. Both throws at once is a defined tie-break, not a fault. PRESS pulses the selected throw through the same path as the hardware.



Figure 5. DIAG · override rockers. Close throw held.

On screen	Meaning
DIAG · override rockers	Header left.
4 / 13	Header right.
Open throw · GPIO10 · T20	Left tile. Open side of both paralleled stations.
Grey pip + OFF HIGH	Open throw not made. Pulled up.
edges 14 · last 00:02:41	Accepted edges since boot, and age of the last open edge.
Close throw · GPIO18 · T8	Right tile.
Green pip + MADE LOW	Close throw is held to ground right now.
edges 11 · last 00:00:03	Close-edge count and the edge three seconds ago.
command model · edge trigger · centre does nothing	Rocker centre is idle. Hold-to-run is not the model.
both throws made · tie-break: OPEN wins	If both throws hit in the same cycle, firmware commands open.
debounce · 25 ms	Switch debounce. Same family as Settings → Triggers.
last accepted command · CLOSE · ignored, already closed	Last edge that passed debounce, and why it did not start a move.
Pin strip: active-low → GND · INPUT_PULLUP · 2 stations	Wiring. Two stations in parallel hide a stuck mate unless this page splits the throws.
10: HIGH · 18: LOW	Live GPIO10 and GPIO18. Stays the real pin during a pulse.
TURN select · PRESS pulse	Highlight open or close. The highlight stops at each end. Extra clicks do not wrap. PRESS injects a short made-edge on that throw, the same command as the physical rocker. It can start a move. Amber pip and SIM while the pulse is live.
HOLD back	Return to the menu. Drops a pulse that has not expired.

6. Motor current load — ACS712-05B

Live amps against the learned envelope for the bin the door is in. Trip is a time-plus-threshold decision, so inrush skip and hold time are on the same page.

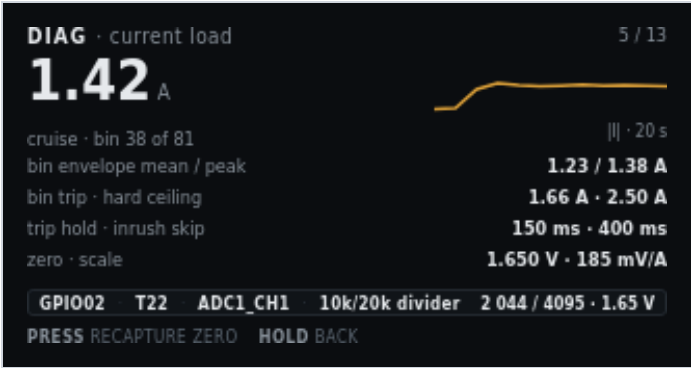


Figure 6. DIAG · current load. Live 1.42 A in cruise.

On screen	Meaning
DIAG · current load	Header left.
5 / 13	Header right.
1.42 A	Hero. Cooked motor current.
cruise · bin 38 of 81	Motion phase and which obstruction bin the door is in.
Sparkline + I · 8 s	Absolute current, newest at the right (dot). About 8 seconds once the strip is full. The trace scrolls left. The step up is the start of cruise.
bin envelope mean / peak · 1.23 / 1.38 A	Learned mean and peak for bin 38.
bin trip · hard ceiling · 1.66 A · 2.50 A	This bin’s trip, then the compile-time ceiling.
trip hold · inrush skip · 150 ms · 400 ms	How long current must stay over trip, and how long after PWM-on the envelope is ignored.
zero · scale · 1.650 V · 185 mV/A	Stage 1a zero and the fixed ACS712-05B scale.
Pin strip: GPIO02 · T22 · ADC1_CH1 · 10k/20k divider	Pin, terminal, ADC channel, Board A divider.
2 044 / 4095 · 1.65 V	Live raw and volts. Near the rest zero, plus load.
PRESS recapture zero	Jump to Calibration stage 1a. Motor must be at rest.
HOLD back	Return to the menu.

7. Motor PWM and direction

Duty and K1 are separate blocks. Duty must reach zero and current must decay before the relay moves, so the blank timers are on this page.

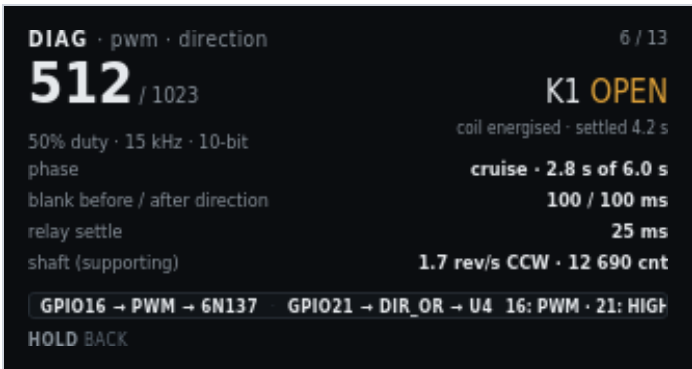


Figure 7. DIAG · pwm · direction. 50% duty, K1 open.

On screen	Meaning
DIAG · pwm · direction	Header left.
6 / 13	Header right.
512 / 1023	Hero. Live duty in 10-bit counts.
50% duty · 15 kHz · 10-bit	Percent, carrier, resolution. Carrier is compile-time. It is not a setting.
K1 OPEN (amber)	Direction relay commanded open. Amber because the coil is energised.
coil energised · settled 4.2 s	K1 has been in this state long enough to be settled.
phase · cruise · 2.8 s of 6.0 s	Profile phase and how much of cruise time has elapsed.
blank before / after direction · 100 / 100 ms	PWM-off windows around a K1 change. Never reverse K1 without this blanking.
relay settle · 25 ms	Wait after K1 before PWM may rise.
shaft (supporting) · 1.7 rev/s CCW · 12 690 cnt	Encoder as a supporting signal on this page. The focused page is PWM.
Pin strip: GPIO16 → PWM → 6N137 · GPIO21 → DIR → U4	PWM path onto Board B, and direction into U4.
16: PWM · 21: HIGH	GPIO16 is toggling. GPIO21 is HIGH for open on this build.
HOLD back	Return to the menu. This page does not command the motor.

8. Manual jog

Owens the motor. Each EC11 detent (or web - / +) queues a fixed run time at jog duty. Same-direction clicks add to that queue. Opposite clicks subtract; when the queue is gone, that click reverses through the normal PWM-off, decay, blank, and K1 settle windows. PRESS while idle cycles the two live settings: duty as a percent (5–25%) and time per detent (200–1000 ms). Shaft encoder, markers, and learned envelope do not start, stop, or abort the move. Current is a live 0–2.50 A graph when the ACS712 is sampling; it is display-only. Travel shows unknown if origin is not trusted.



Figure 8. DIAG · jog. Timed queue, duty as percent, live 0–2.50 A graph. PRESS while idle edits duty and step.

On screen	Meaning
DIAG · jog	Header left.
7 / 13	Header right.
MOTOR LIVE — JOG	Amber banner title. This page owns the motor. Hall and rocker cannot preempt it.
EC11 only — opposite blanks before reverse	Banner detail. Only the knob or web keypad commands this page. A reverse never hot-switches K1.
OPEN / CLOSED	Travel ends.
Blue fill to 62%, yellow thumb, two marker ticks	Same travel widget as Main when position is trusted. Door is at 62% of span. Missing encoder: unknown and -.
62% · 12 690 cnt	Share of span and counts from origin, if the shaft magnet is good.
jog duty · direction · 9 % · CLOSE	Jog PWM as a percent of 10-bit full scale (5–25%). K1 for this jog. PRESS while idle highlights this row; TURN changes it.
step · remaining · 400 ms · 800 ms	Time added per detent (200–1000 ms, 50 ms steps) and how much powered time is still queued. PRESS again highlights this row. Queue caps at 4 s.
Live amp strip · 0.41 A · 0–2.50 scale	ACS712 history, last ~8 s. Red line is the 2.50 A ceiling. Not used to run or stop the jog.
Pin strip: GPIO16 · GPIO21 · T5 / T6	PWM and direction pins, Board A motor terminals.
12 690 cnt or enc --	Shaft counts when the magnet is good. Display-only.
TURN jog or TURN duty / TURN step	Default: CW queues close, CCW queues open. Same way extends. Opposite shortens, then blanks and reverses. With a setting highlighted, TURN changes that value.
PRESS stop or PRESS edit / PRESS next	While moving: drop PWM. While idle: cycle Jog → duty % → step ms → Jog. Writes the value when you leave a field.
HOLD exit	Stop if moving, save a pending edit, then return to the menu.

9. Obstruction test

Plastic bottle between the doors. Same envelope rules as Main. Does not write the learned table. Learn runs assume a clear path. This test is not a substitute for them.



Figure 9. DIAG · obstruction test. Close trip, reversed, opening.

On screen	Meaning
DIAG · obstruction test	Header left.
8 / 13	Header right.
TRIPPED — REVERSED	Amber banner title. Close obstruction fired. Reverse is in progress or just finished.
Close obstruction at bin 41 · 2.11 A vs trip 1.65 A	Where and how far over trip.
test load · plastic bottle between doors	Specified test article. Not a body part, not a tool.
expected close / open · reverse once / fault first trip	Close reverses once. Open faults on the first trip. Same as Main.
result this run · reversed, then opened to marker	What this test actually did.
table writes · hard ceiling · disabled · 2.50 A armed	A deliberate obstruction cannot be taught as load. The ceiling stays on.
Pin strip: GPIO02 current · GPIO16 pwm · GPIO21 dir	Signals this test uses.
1.02 A · opening	Live current and direction after reverse.
PRESS run again	Start another close into the bottle.
HOLD exit	Cancel if live, then return to the menu.

10. Obstruction table — close direction

Learned envelope as a curve. Breakaway at the open end and seal loading at the close end are the shape you want. A bump in the middle is a bind. The grey block past the close marker is travel the learn runs never enter, because snug is a separate window.

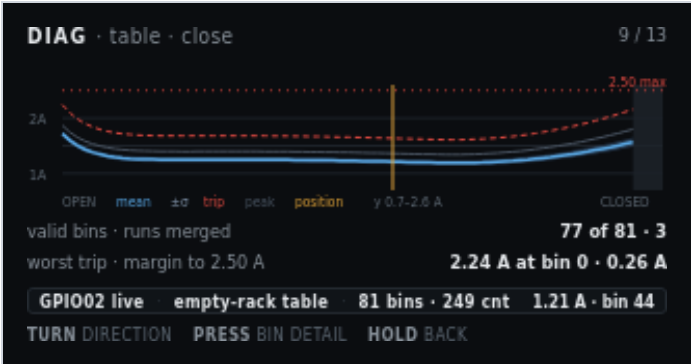


Figure 10. DIAG · table · close. Empty-rack table, live at bin 44.

On screen	Meaning
DIAG · table · close	Header left. Close-direction envelope.
9 / 13	Header right. Menu item 9 is this pair of graphs.
1A / 2A on the Y axis	Current scale. The window is 0.7–2.6 A so the curve is readable on 170 pixels.
Dashed red line + 2.50 max	Compile-time hard ceiling. Not a learned value.
Blue mean curve	Merged mean current per bin from the learn runs.
Grey band $\pm\sigma$	One-sigma band around the mean.
Dashed red trip curve	Per-bin trip used on Main.
Thin grey peak curve	Highest sample merged into that bin.
Vertical amber position line	Live door position on this graph.
Dark block near CLOSED	Bins past the close marker. Learn runs never visit them. Snug is a separate window.
Legend: OPEN · mean · $\pm\sigma$ · trip · peak · position · y 0.7–2.6 A · CLOSED	X ends and the plotted series. Y window is stated so the clipped axis is not mistaken for 0–2.50 A.
valid bins · runs merged · 77 of 81 · 3	Bins with data, and how many strokes were merged.
worst trip · margin to 2.50 A · 2.24 A at bin 0 · 0.26 A	Highest trip on this table (breakaway at the open end) and room left to the ceiling.
Pin strip: GPI002 live · empty-rack table · 81 bins · 249 cnt	Which table is drawn, bin count, and bin width in counts.
1.21 A · bin 44	Live current and the bin under the position line.
TURN direction	Swap to the open-direction graph.
PRESS bin detail	Numeric values for the bin under the cursor (not a separate mockup in this set).
HOLD back	Return to the menu.

11. Obstruction table — open direction

Same table, other direction. Breakaway is now at the closed end, where the door pulls out of the seal. The unvisited block moves to the open end. Two independent tables are why both views exist.

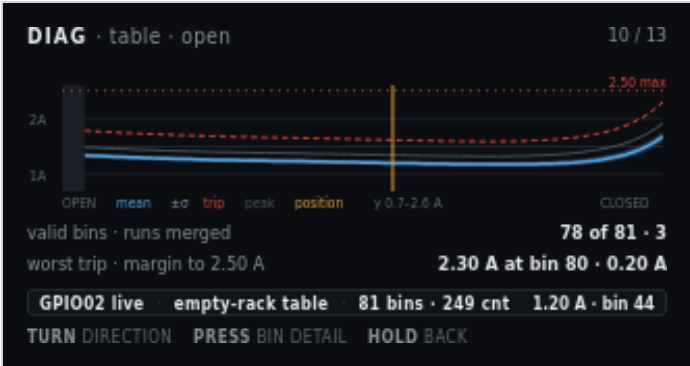


Figure 11. DIAG · table · open. Breakaway at the closed end.

On screen	Meaning
DIAG · table · open	Header left.
10 / 13	Header right.
Graph, Y labels, 2.50 max, legend	Same series as the close graph. Shape differs: peak at the closed end.
Dark block near OPEN	Bins before the open marker. Open-direction learn runs never visit them.
valid bins · runs merged · 78 of 81 · 3	Open-direction bins with data (bins at and after the open marker) and merged run count.
worst trip · margin to 2.50 A · 2.30 A at bin 80 · 0.20 A	Highest open-direction trip (breakaway at the closed end) and room left to the ceiling.
Pin strip: GPIO002 live · empty-rack table · 81 bins · 249 cnt	Same table family as the close view.
1.20 A · bin 44	Live current and the bin under the position line, read against the open envelope.
TURN direction · PRESS bin detail · HOLD back	Same gestures as the close graph.

12. EC11 knob and button

An input with no other home. Detent scale is bench-calibrated, so raw quadrature and the scaled count sit side by side.

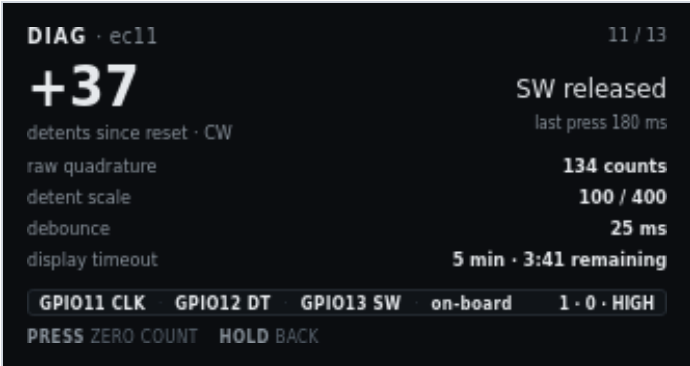


Figure 12. DIAG · ec11. Detent scale 100 / 400. Display timeout 5 min.

On screen	Meaning
DIAG · ec11	Header left.
11 / 13	Header right.
+37	Hero. Scaled detents since reset. Sign is CW positive on this shot.
detents since reset · CW	What the hero counts, and last turn sense.
SW released	Knob switch is up.
last press 180 ms	Duration of the last PRESS (under the HOLD threshold).
raw quadrature · 134 counts	Unscaled CLK/DT edges. Compare with the detent hero to check the scale.
detent scale · 100 / 400	Quadrature-to-detent fraction from ec11_calibration.h (kEc11DetentScaleNumerator / kEc11DetentScaleDenominator). 100/400 is one detent per four edges.
debounce · 25 ms	SW debounce.
display timeout · 5 min · 3:41 remaining	Idle blanking setting and time left until GPIO38 backlight and GPIO15 panel power drop. Default is 5 minutes.
Pin strip: GPIO11 CLK · GPIO12 DT · GPIO13 SW · on-board	EC11 on the T-Display-S3, not a Board A terminal.
1 · 0 · HIGH	Live CLK, DT, SW. SW HIGH is released with pull-up.
PRESS zero count	Reset the detent hero to 0. Does not change calibration constants.
HOLD back	Return to the menu.

13. Network and OTA

Setup surface. The steel lid can block the onboard antenna, so that note is on the screen, not only in the build docs. When SoftAP is serving, this page shows the join password in clear (default reserved). The AP name defaults to PrivateReserve. After join, open `http://192.168.4.1/`. PRESS rejoin retries the stored STA SSID. To drop that SSID, use Settings → Service → Clear station Wi-Fi, then reboot.

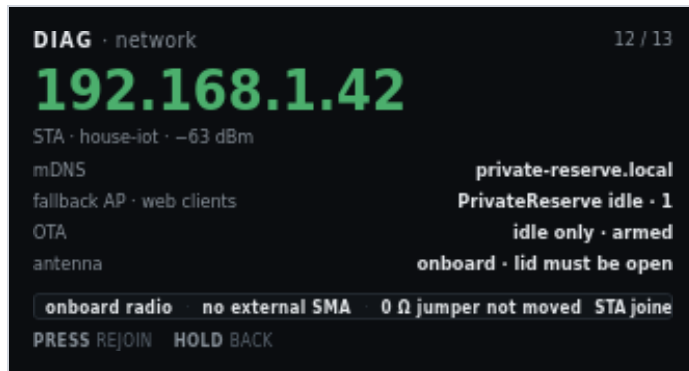


Figure 13. DIAG · network. STA joined, OTA idle-only.

On screen	Meaning
DIAG · network	Header left.
12 / 13	Header right.
192.168.1.42 (green)	Hero. Station address when STA joined.
STA · house-iot · -63 dBm	Role, stored SSID, RSSI.
mDNS · private-reserve.local	mDNS name for HTTP on the joined LAN.
fallback AP · web clients · PrivateReserve idle · 1	SoftAP name, idle because STA joined, and one HTTP client. If SoftAP is the live link, this row is join password and the WPA2 PSK (default reserved) instead.
OTA · idle only · armed	OTA is allowed only while the motor is idle. Armed means the endpoint is listening.
antenna · onboard · lid must be open	No external SMA. Close the steel lid and RSSI can collapse.
Pin strip: onboard radio · no external SMA · 0 Ω jumper not moved	Antenna path as shipped. Do not move the 0 Ω jumper unless you fit an external antenna.
STA joined	Live radio role.
PRESS rejoin	Retry STA join with the stored SSID. If join fails, SoftAP starts. This does not clear the stored password. To wipe station credentials, use Settings → Service → Clear station Wi-Fi, then reboot from that result screen or from Reboot controller.
HOLD back	Return to the menu.

14. Stored calibration

Answers whether this controller may run automatically, and why not. Stale entries name the value that invalidated them.

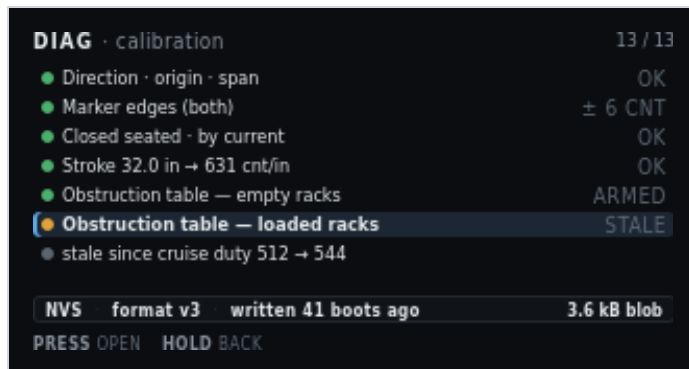


Figure 14. DIAG · calibration. Loaded table stale after cruise duty 512 → 544.

On screen	Meaning
DIAG · calibration	Header left.
13 / 13	Header right. Last detail screen.
Green pip + Direction · origin · span + OK	Stages 2–4 stored and valid.
Green pip + Marker edges (both) + ± 6 CNT	Open/close probe spread still inside tolerance. The value column shows the open-side figure on this shot.
Green pip + Closed seated · by current + OK	Stage 5 method still current-rise.
Green pip + Stroke 32.0 in → 631 cnt/in + OK	Stage 6 still matches the stored span.
Green pip + Obstruction table – empty racks + ARMED	Empty envelope is in service.
Amber pip + highlight + Obstruction table – loaded racks + STALE	Selected. Loaded table is not usable.
Grey pip + stale since cruise duty 512 → 544	Why. A motion-profile change invalidates both tables. This row names the field that did it.
Pin strip: NVS · format v3 · written 41 boots ago	Where the blob lives, format, age in boots (no RTC).
3.6 kB blob	Stored size.
PRESS open	Jump to the matching Calibration stage for the highlighted row.
HOLD back	Return to the menu.

15. Related

- [Software overview](#) — full web UI, figure 1
- [Settings](#) — clear station Wi-Fi and reboot
- [Board A](#) — T7, T8, T9/T10, T20–T23

This page is a chapter of the Software User Manual.